

EECS 6895 • ADVANCED BIG DATA & AI • FINAL PROJECT

ViSQL

*A Vision-Augmented Autonomous Data Scientist
for End-to-End Natural-Language Analytics*

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Analysts spend their time on the wrong work.

THE REPEATING PATTERN

1

Stage 1 — SQL plumbing

Schema lookups, joins, dialect quirks, debugging errors.

2

Stage 2 — Visualization

Reshape data, pick chart types, copy a teammate's aesthetic.

3

Stage 3 — Narrative

Translate findings into a report stakeholders will read.

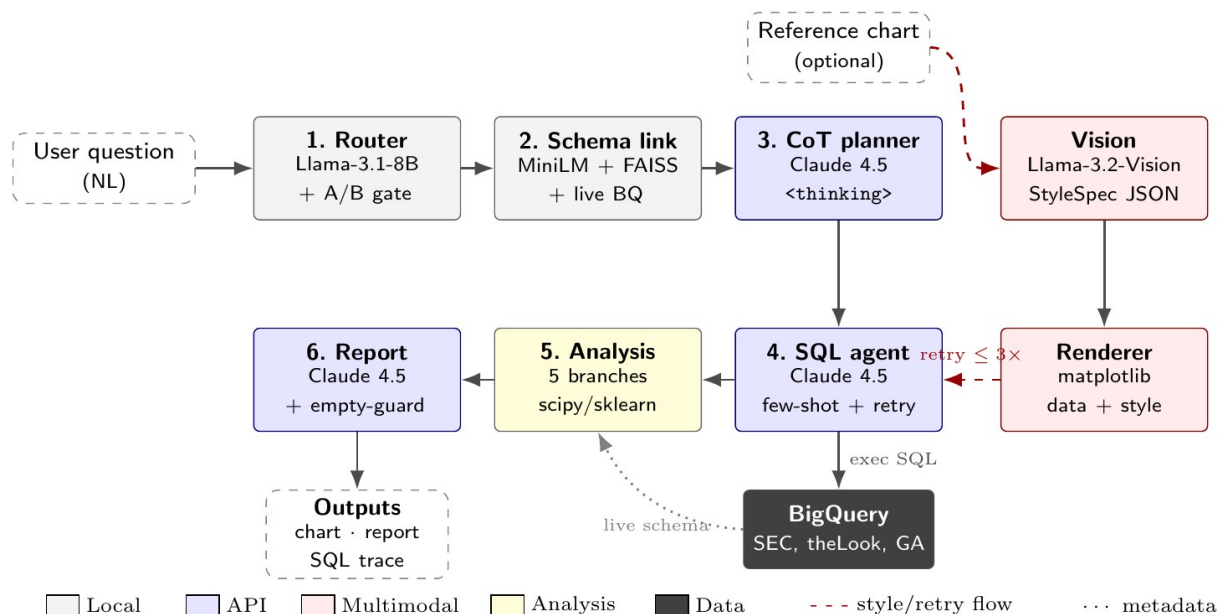
THE GAP

LLM tools today either...

**Write the SQL.
Or summarize results.
Or generate a chart.**

Nobody does the whole loop the way an analyst actually does — with style imitation, statistical rigor, and grounded narrative.

Six-stage hybrid pipeline.



Local Llama (router, vision) + Claude API (planner, SQL, report). Vision is an optional parallel subsystem.

Three architectural guardrails.



Strict A/B test gate

ab_test fires ONLY if the question has experiment-language OR the schema has a variant column. Observational comparisons are demoted to single_chart — preventing causal claims the data can't support.



Live schema introspection

Schemas fetched at runtime from INFORMATION_SCHEMA. Eliminates schema drift. Pre-execution table-existence check catches hallucinated tables before the BigQuery round-trip.



Empty-data guard

If SQL returns 0 rows, the report writer refuses to narrate. Emits the SQL + diagnostic instead of fabricating findings — a failure mode we observed repeatedly in early iterations.

DEMO

Live walk-through of the pipeline on three queries.



Standard question → styled chart + grounded report



A/B gate catches an observational "is X different from Y" question



Reference-chart upload → matplotlib renders user's data in that style

The A/B test gate, in two queries.



REAL EXPERIMENT • passes gate

"For the checkout_redesign_2024 A/B test, did treatment lift conversion vs control?"

Route **ab_test**

Stat result

$\chi^2 = 412.7$, $p < .001$, lift = +8.1pp

Significant. Cited in narrative report.



OBSERVATIONAL • gated

"Is conversion significantly different between mobile and desktop users?"

Route (LLM proposed → gated)

ab_test → **single_chart**

Why

No experiment-language. No variant column.

Causal claims would be unsupportable.

Four suites. Real numbers.



Router (macro-F1)

0.92

60 hand-labeled • 5 task classes
Gate: 8/10 on adversarial probes



Style imitation

ΔE 7.4

12 reference charts • CIELAB self-consistency
Below JND threshold of 10



Reports (LLM judge)

4.1/5

6 questions × 4 axes
Coverage 0.78 • empty-guard verified



SQL on Spider

0.74

200-ex Spider 1.0 dev slice
LoRA SQLCoder baseline: also ~0.74

Self-correction is doing real work.



MARGINAL LIFT FROM RETRY

+6 pp

first-attempt 0.68 → final 0.74

The retry loop transforms ~6 percentage points of failure into success — invisible without per-attempt instrumentation.

HOW THE RETRY LOOP BEHAVES

Metric	Value
Retry-rate (≥ 1 retry)	17%
Recovery-rate	55%
First-attempt exec-acc	0.68
Final exec-acc	0.74

Error-class breakdown: *missing_object*, *syntax_error*, *wildcard_or_dataset_qualification*, *type_error*.

In all four, pasting the error back in-context lets the model fix the same mistake on the next pass — the failure mode is local.

The main lesson of this project.

The most damaging LLM failures are not reasoning errors — they are routing errors.

Fixing them at the routing layer is dramatically more reliable than asking a downstream component to suppress them.



Empty-data guard

Refuses to narrate when SQL returns 0 rows. No more confidently-narrated wrong answers.



A/B test gate

Prevents causal language from ever being requested — strictly stronger than asking the writer to refuse it.



SQL retry loop

Errors fed back in-context; +6 pp lift. Local failures don't propagate downstream.

SUMMARY

Closing the analyst loop.

WHAT VISQL DELIVERS

- End-to-end NL → SQL → chart → report pipeline
- Three guardrails: A/B gate, schema introspection, empty-data guard
- Multimodal style imitation from reference charts (ΔE 7.4)
- 0.74 Spider exec-acc, with +6 pp from self-correction
- Hybrid local Llama + Claude API, ~3.6K LOC, 14 modules

WHAT I'D BUILD NEXT

Three priorities.

1

BIRD-bench eval

Spider 1.0 too easy by 2026. BIRD adds dirty data + ambiguous joins.

2

More A/B gate coverage

Multi-arm, sequential, ambiguous phrasings. Externally-labeled set.

3

Planner as first-class

Re-feed <thinking> as constraints into a self-debugging SQL loop.

Thank you. · Questions?